



TRAX IO

AI Inventory Optimization for Airline Spares

A multi-tenant, agentic optimizer layered on Trax eMRO — continuously tuning re-order point, order quantity, safety stock and max per part, per location, under governed autonomy.

Static stock levels, hand-tuned once a year, in a live fleet

eMRO's PN_INVENTORY_LEVEL holds one fixed (ROP, EOQ, Safety Stock, Max) per part, per location — set by hand, reviewed maybe annually. Meanwhile demand, lead times, wash rates and criticality drift every day. Get a number wrong in either direction and it costs real money.

The tuning is manual, coarse, and always out of date.



Set too low → AOG risk

- Grounded aircraft and dispatch-reliability hits
- Emergency expedites, AOG fees, borrow / loan costs
- Cascading schedule disruption across the network



Set too high → trapped capital

- Millions in slow-moving and obsolete stock
- Shelf-life write-offs on time-limited parts
- Working capital locked up instead of deployed

An always-on optimizer that plans every part, every location

Trax IO replaces the static PN_INVENTORY_LEVEL values with policy-driven ones — recomputed per PN × Location across every tenant's catalog, continuously.



Continuous recompute

Nightly full-fleet, plus real-time recompute on flight, stock and AD/SB events.



Regime-routed forecasting

The right statistical model for each demand class — from ultra-rare to high-volume.



Deterministic policy engine

The LLM informs; deterministic Python decides and writes. Planners sign it off.



Governed autonomy

Tiered A/B/C writes, hard guardrails, full audit and 90-day rollback.

Three forces converge in 2026

1



Agentic AI is production-ready

Generative-AI-native agents on AWS Bedrock AgentCore are operationally mature. The same bet two years ago would have been research-grade.

2



Carriers are actively buying

Tier-1 carriers are scoping AI inventory products now. The first credible MRO-native vendor wins multi-year contracts.

3

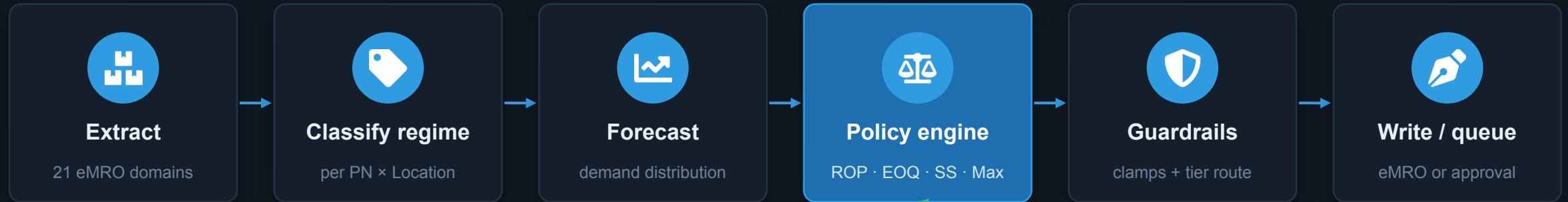


Trax already owns the data

The eMRO catalog, demand history, vendor terms, utilization and criticality. Competitors must integrate; we ship one that already knows the data model.

Two cadences on one pipeline: nightly full-fleet, real-time on events

① NIGHTLY — every active part, every location



② REAL-TIME — hot-parts recompute (hourly)

eMRO outbound events (stock moved · flight completed · removal recorded · AD/SB issued) fire Step Functions to re-solve only the affected part-locations — feeding the same policy engine and guardrails.

The recommendation is never a black box: every output carries a provenance record — model, distribution parameters, service-level target, binding constraints, and delta vs. the current value.

Four levels, from five inputs, with full provenance

FIVE INPUTS

 Demand distribution

 Lead-time distribution

 Service-level target

 Cost structure

 Hard constraints





Policy Engine
deterministic Python — no LLM in the write path



FOUR LEVELS

ROP
Re-Order Point

EOQ
Economic Order Qty

SS
Safety Stock

Max
Maximum level

SERVICE-LEVEL TARGETS BY ESSENTIALITY TIER

99.5%	98%	95%	92%	90%
AOG / NO-GO	GO-IF	Dispatch-critical	Routine	Consumable

Tiered autonomy: aggressive on the long tail, conservative on AOG



TIER

Advisor only

- Recommendation lands in the eMRO review queue
- Planner approves → Writeback Agent writes
- AOG / flight-safety · unit cost $\geq$ \$10K · $\Delta > 25\%$
- First 90 days of every deployment

Highest-risk parts



TIER

Bounded autonomy

- Agent writes within policy bands
- Out-of-band → escalates to Tier A
- GO-IF / dispatch-critical · Δ within $\pm 15\%$
- 14-day notification + one-click rollback

The middle band



TIER

Autonomous + digest

- Agent writes; planner sees a weekly digest
- Routine expendables / consumables · $\Delta \pm 40\%$
- Unit cost $<$ \$500 · high-volume regime
- Where most of the savings live

60–70% of the catalog



Hard guardrails never bypassed. Single-write delta capped at 100% even in Tier C · shelf-life / hazmat / tool-control clamps · an active AOG in the last 72h forces Tier A · per-tenant kill switch reverts to shadow mode in ≤ 60 seconds.

Safety engineering: guardrails, kill switches, incident response



Bounded writes

Single-write delta capped at 100% — even in Tier C. Floors enforced: $SS \geq 0$ · $ROP \geq SS$ · $Max \geq ROP + EOQ$ · $EOQ \geq MinOQ$.



Two kill switches

Per-tenant revert to shadow mode in $\leq 60s$; a global Trax kill switch stops all tenants in ≤ 5 min. 90-day rollback window, never zero.



Shelf-life & hazmat clamps

$Max \times avg \text{ daily demand} \leq 0.6 \times \text{shelf-life days}$. Hazmat / tool-control parts cannot more than double per write cycle.



Five incident runbooks

Model garbage · writeback failure · event-lane stall · data leakage · prompt injection — each with a defined response and a 30-day action SLA.



Deferral & AOG override

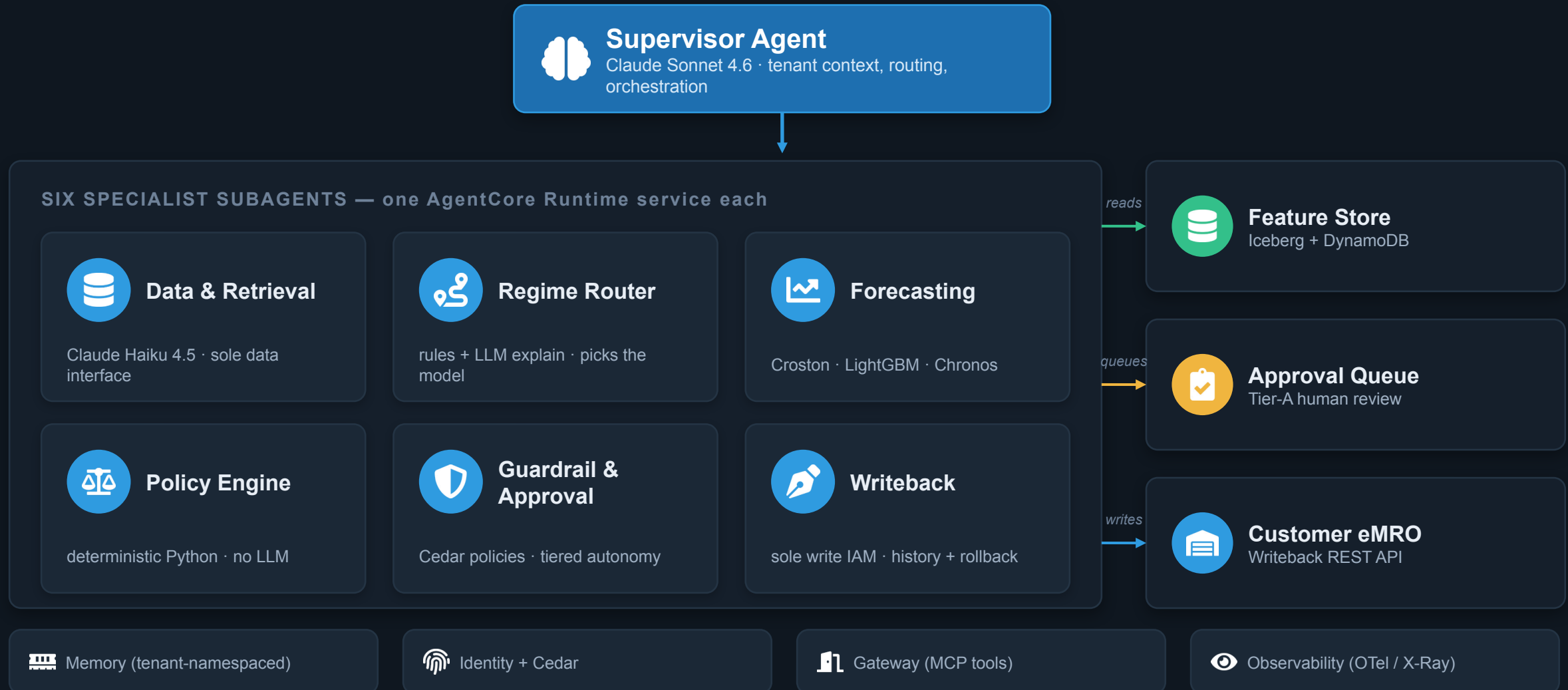
Open POs/ROs that would breach Max → defer one cycle. An active AOG in the last 72h forces Tier A regardless of other criteria.



Red-team suite

Every model & agent release is regression-tested against adversarial scenarios: wash-rate spike, lead-time collapse, AD shock, seasonal inversion, phantom demand.

One supervisor, six specialists, on AWS Bedrock AgentCore



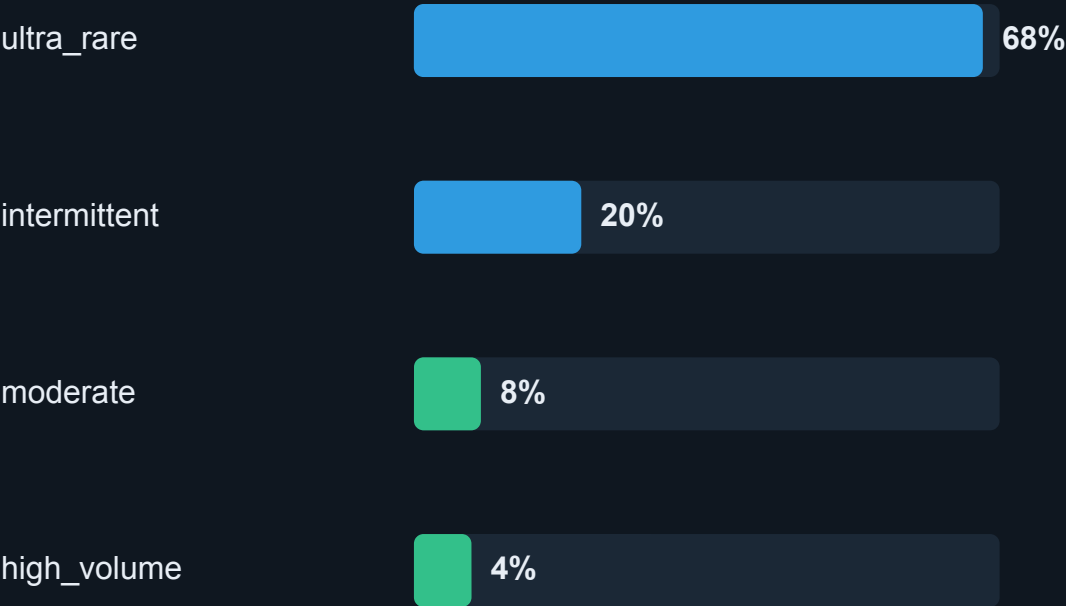
Six specialists, each independently versioned

Specialist	Role	Model / engine	Access
Data & Retrieval	Sole interface to feature store, extracts & event lane — every other agent reads through it	Claude Haiku 4.5	read
Regime Router	Classifies each key's demand regime and selects the forecast model	rules + Sonnet 4.6 explain	read
Forecasting	Hosts three regime models behind one interface; delegates inference to SageMaker	Croston · LightGBM · Chronos	read
Policy Engine	Turns inputs into (ROP, EOQ, SS, Max) with provenance — what planners sign off	deterministic Python	read
Guardrail & Approval	Enforces tiered autonomy via Cedar; routes out-of-band recs to the approval queue	Cedar + Sonnet 4.6	read
Writeback	Writes approved changes, logs history, handles rollback	deterministic Python	WRITE

The Supervisor (Sonnet 4.6) owns tenant context and routing — it never does domain work. **Phases 2–6 each add one specialist without changing the Supervisor contract.**

The right model for each demand regime

SHARE OF CATALOG BY COUNT



~68% of parts are ultra-rare — the long tail is the hard, valuable problem.

ultra_rare

< 6 removals / 24 mo

Empirical-Bayes Gamma-Poisson

peer-part priors (same ATA, criticality, fleet)

intermittent

6 – 24 removals

Croston / TSB / SBA

classical intermittent, via statsforecast

moderate

25 – 200 removals

LightGBM + causal covariates

wash rate, utilization, criticality, ATA

high_volume

200+ removals

LightGBM + foundation ensemble

blended, weighted by rolling MAPE

Every model returns a full demand distribution, not a point forecast. *Re-classified on a $\pm 20\%$ hysteresis band; Chronos / Moirai challengers run in shadow.*

No silent model swaps. Ever.



Champion / challenger

- Scored nightly on a rolling 60-day holdout
- Three metrics: weighted MAPE · fill-rate vs target · total cost
- Challenger promoted only after beating champion on all three for 45 consecutive days
- Always preceded by a planner-visible notice



Federated peer priors

- Challengers train on de-identified PN-level features
- Wash rate, causal intensity, criticality, ATA, fleet type
- No tenant IDs, part numbers or vendors cross the boundary
- Separate AWS account, no read access to tenant PII



Distributions, not points

- Every model emits mean, variance and tail percentiles
- Feeds service-level-constrained cost minimization
- Peer-median benchmark ships as a v2 premium feature
- Fallback to current static level when confidence is low

The Feature Store — the single most load-bearing item



Offline — Glue + Apache Iceberg

11 versioned feature groups, recomputed nightly

- Partitioned by tenant, Iceberg time-travel enabled
- Any historical recommendation is reproducible
- SOC 2, audit, and planner-trust multiplier



Online — Amazon DynamoDB

thin (tenant, pn, location) layer

- Sub-10 ms reads for event-triggered inference
- Populated nightly by Glue, updated by the event lane
- Right-sized: the read pattern doesn't justify SageMaker FS

ELEVEN FEATURE GROUPS

green = added in Wave 4 (demand-side open requisition lines)

part_attributes

criticality

stock_position

current_policy

vendor_economics

lead_time_distribution

demand_history

interchangeable_graph

location_graph

open_orders_snapshot

requisition_snapshot

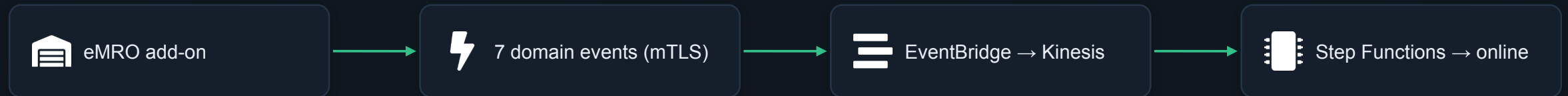
How eMRO data reaches the agent

Nightly extract — the workhorse



Signed utility runs on the customer's own scheduler · 21 canonical SQL domains

Event lane — real-time



flight_completed · stock_moved · wo_scheduled · vendor_price_changed · plan_published · removal_recorded · eo_published

Optional CDC — tier-1 upsell



Hourly hot-parts without waiting for the Event Publisher release

One write path. Every change reversible.



Writeback is the ONLY agent with write access to eMRO.

Every other specialist is strictly read-only. One narrowly-scoped IAM role holds `PN_INVENTORY_LEVEL:Write`.



Provenance ledger

Transactional writes to `PN_INVENTORY_LEVEL`, logged to `PN_INVENTORY_LEVEL_HISTORY`: old → new, agent, version, forecast id, tier, timestamp.



90-day rollback

Every write reversible for 90 days (configurable, never zero). Bulk rollback requires a second planner confirmation.



Shadow mode

During onboarding every would-be write is logged as SHADOWED and nothing is applied — proof before power.



Dual-book audit

Mirrored to immutable S3 (Object Lock). Neither party can unilaterally rewrite history.

Trax IO Review — one workspace for the planner



Overview

Portfolio KPIs, health mix, ATA risk, priority actions



Part Drill-Down

Attributes, stock, lead time, demand, history



Workbench

The server-paged core approval loop



AI Recommendations

Explainable rec → reason → action cards



Forecast & Service Levels

Projected demand vs. target fill rate



What-If Scenarios

Live-solved SL / budget / TAT sliders



Data & Connections

Honest 13-feed connection-status table

The provenance invariant

Every number carries its lineage — source, system of record, freshness, coverage, confidence. A value can't render without it.

React 18 + TypeScript + Tailwind · one frontend over a FastAPI backend-for-frontend · WCAG 2.1 AA.

The approval loop, built for 40k-SKU catalogs

Workbench

Bulk approve

Export CSV

PartLocCurrent → ProposedΔTier

9821-4	JFK	ROP 4 → 6	+50%	B
3340-1	LHR	Max 12 → 10	-17%	C
771-A2	SIN	SS 2 → 5	+150%	A
5120-9	DFW	EOQ 8 → 9	+12%	C
6002-7	AMS	ROP 1 → 2	+100%	B

Server-paged (200/page) · filter + sort · accept / reject / defer · per-row history

Overview

58.9K

Planning-active keys

\$419M

Open pipeline

1,082

Shortage parts

11.1K

Open recs

PORTFOLIO HEALTH MIX

Healthy 55%

Watch 28%

At-risk 17%

Drill-in panels on every card · illustrative figures

From a single part to the whole what-if



Part Drill-Down

- Attributes, description & criticality
- Stock breakdown by location
- Lead-time view: promised vs. actual
- Open orders & 24-month demand series
- Writeback history timeline + rollback



What-If Scenarios

- Live-solved service-level / budget / TAT sliders
- Cost-service efficient frontier
- Save, compare and commit scenarios
- See the trade-off before you write it



Data & Connections

- Honest 13-feed connection-status table
- What's live vs. what's stubbed
- Freshness and coverage per feed
- No feed pretends to be more than it is

The Business Value Report — the renewal engine

PROJECTED SAVINGS vs. PRE-AGENT BASELINE

Holding cost



Ordering cost



Stockout risk



Projected-only attribution — each component with its disclosed inputs (e.g. count of parts where EOQ was non-positive and ordering was skipped).



Tier posture

Distribution of recommendations across Tier A / B / C.



Governance

Approval, override and deferral rates by planner; kill-switch events.



Forward look

Projected 12-month savings, with the extrapolation method disclosed.



Methodology & coverage

Discloses keys valued of total portfolio keys — never implies full coverage.

Schema-locked BvrReport 1.1.0 · JSON / HTML / PDF · auto-generated monthly · CFO-facing.

Tenant isolation at four layers



Contract

TenantContext propagation · per-tenant signed extract utilities

1



Agent

Specialist._assert_tenant_match validated on every decision

2



Data

Feature Store namespace + a dedicated KMS key per tenant

3



Infrastructure

Per-tenant AWS CDK stacks

4

ENFORCED BY



Per-tenant KMS keys

envelope encryption, annual rotation



Cedar authorization

every tool call carries identity



IAM role boundaries

least privilege per specialist



One TraxAi account

isolation without account sprawl

SOC 2 Type II from day one — not bolted on



Immutable audit

Every ingress and writeback mirrored to S3 Object Lock (Compliance mode, 7-year retention). Dual-book — neither party can rewrite it.



CloudTrail Lake

Every access attempt logged, 7-year retention. Quarterly control tests automated via AWS Config + Audit Manager.



Provenance & lineage

Every change and every model version fully tracked: dataset hash, training-job hash, approver, champion/challenger eval record.



Kill switch + incident response

Five incident-class runbooks (model garbage, writeback failure, event stall, data leakage, prompt injection) with a 30-day action SLA.



Red-team suite

Every release regression-tested against adversarial scenarios before it ships.



EU AI Act alignment

Model Registry lineage compatible with high-risk-system documentation requirements.

HOW IT'S BUILT

Built on AWS, per tenant, as code



Agents

Bedrock AgentCore · Strands · Cedar · per-tenant CDK



Models

Claude Sonnet 4.6 · Claude Haiku 4.5 · no LLM in policy / writeback



Data

Glue + Apache Iceberg · DynamoDB · S3 (+ Object Lock) · optional DMS



ML

SageMaker · statsforecast · LightGBM · scipy
EB · Chronos / Moirai



Events

EventBridge · Kinesis · Step Functions



Observability & security

OTel → X-Ray · CloudWatch · Grafana ·
CloudTrail Lake · Audit Manager · KMS

Frontend React 18 + TypeScript + Tailwind · one frontend · FastAPI backend-for-frontend · WeasyPrint for the BVR PDF

What actually runs today

Waves 0–2

coded & fully tested

58.9K

live eMRO keys in local Docker

266

agent-spine / BFF tests passing

288

web (Vitest) tests passing



Proven end-to-end in plain Python. A fully-tested local / Docker reference implementation runs the whole pipeline — extract → forecast → policy → guardrail → writeback → BFF → UI — including the forecasting slices (statistical, gradient-boosted, empirical-Bayes) and the #8 Business Value Report.



No AWS deployment yet. The reference implementation proves the logic; a contract-tested fake_emro stands in for the Oracle write path, and the AWS CDK is coded toward — not deployed. This is a validated blueprint, honestly labelled.

THE PLAN

10 sub-projects across four waves

Wave	Weeks	Sub-projects	Outcome
0 · Foundation	0–6	Extract Utility · Feature Store · Observability / SOC 2	Data flowing; audit plane live
1 · Core agent	6–16	Agent Spine · Forecasting & Policy · Event Publisher · Writeback REST	Full pipeline producing real recommendations
2 · UI + value	12–20	Planner UI · Business Value Report Pipeline	Customer-facing surface + value reporting
3 · Go-live	20–24	Tenant Onboarding	Lighthouse exits shadow → Tier B/C writes live

KEY MILESTONES



Priced per part under management; the BVR renews it

\$25–60K

per tenant / month

Trax IO Core, by catalog size & tier

70–85%

gross margin

even at low scale

\$3–8K

run cost / tenant / mo

Bedrock + SageMaker in shadow

LAYERED SKUs — EACH INDEPENDENTLY COMMERCIALIZABLE





The monthly Business Value Report is the contract-renewal engine. It quantifies dollars saved per tenant. By v3 the conversation shifts from “dollars saved in inventory” to “AOG hours prevented” — a qualitatively higher price tier.

One platform; a new specialist each phase



Each phase adds one specialist without re-architecting the platform. **The commercial story moves from dollars saved to AOG hours prevented.**

From static thresholds to a **governed, learning agent.**



Continuous



Explainable



Governed



Multi-tenant

Trax IO recomputes every part, every location, every day — with the audit trail, guardrails and rollback that let a planner trust an agent with the catalog.